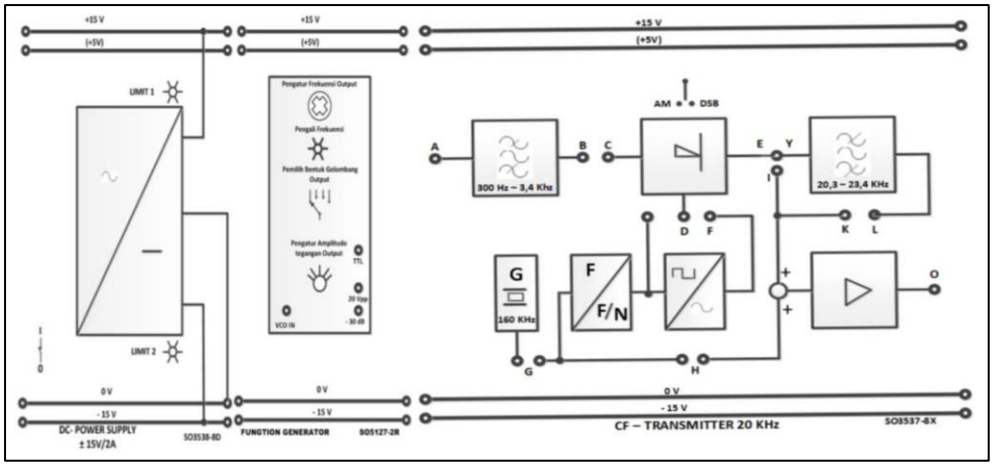
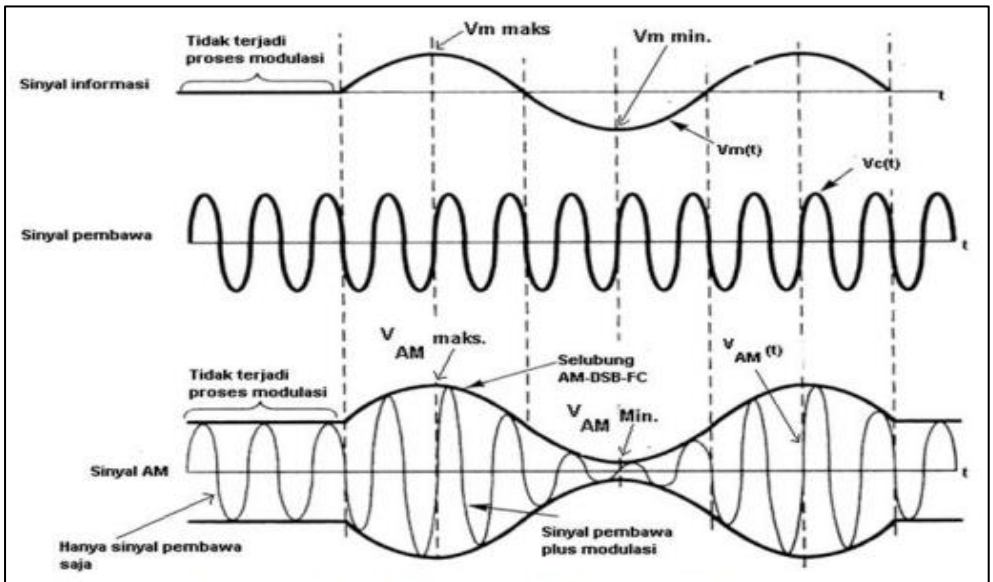
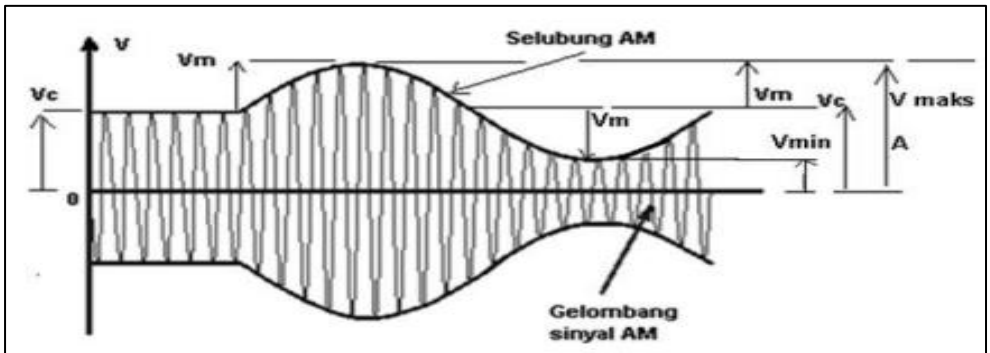
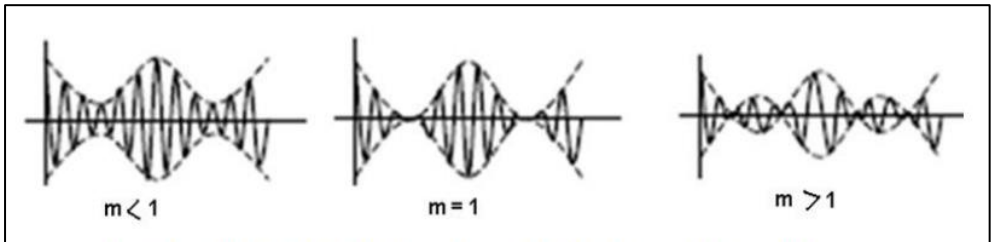




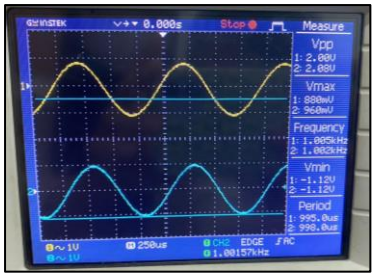
Gambar 3. Bentuk Sinyal Input-output Modulator AM-DSB-FC

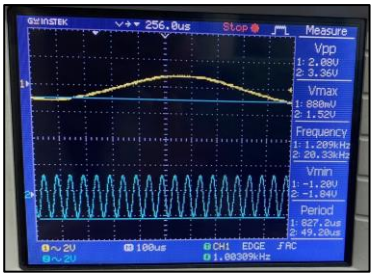
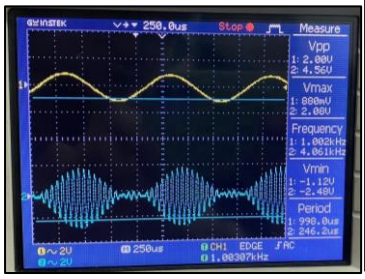
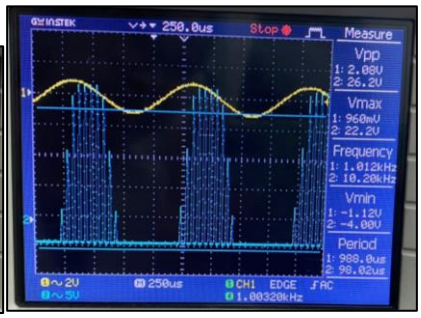
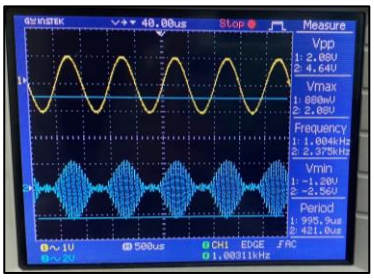
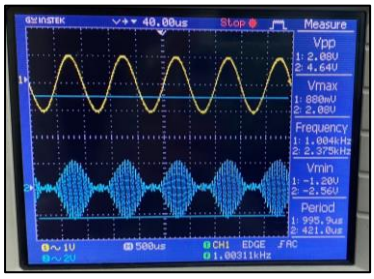
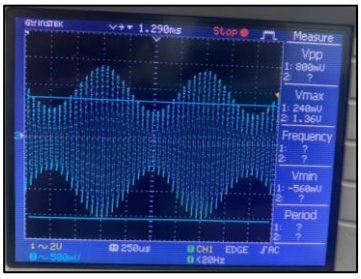
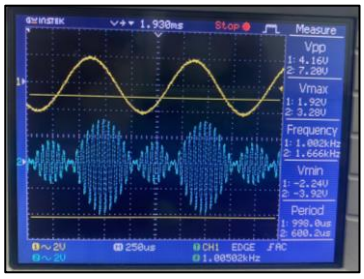
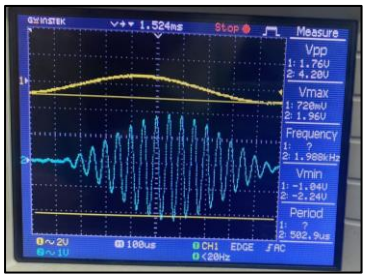
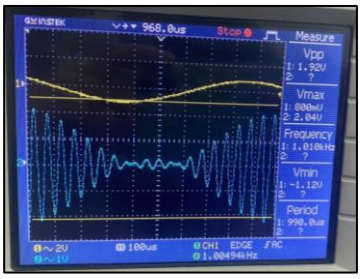
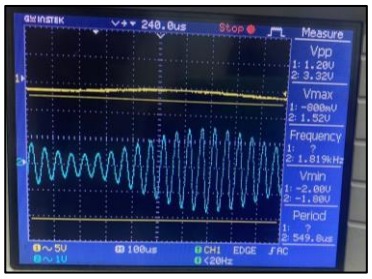
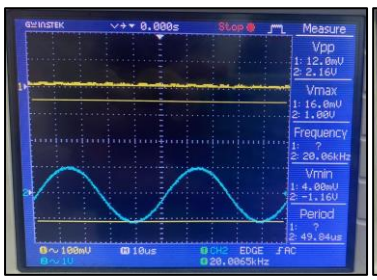
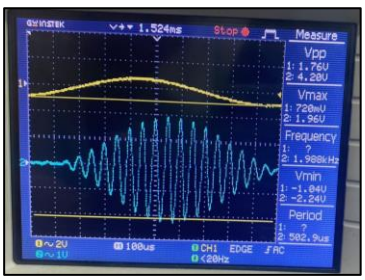


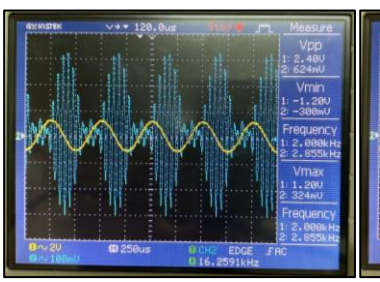
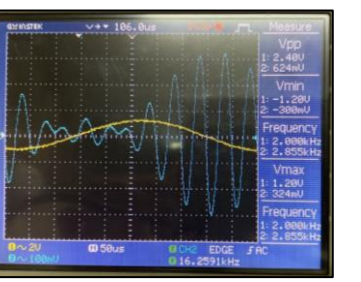
Gambar 4. Indeks Modulasi

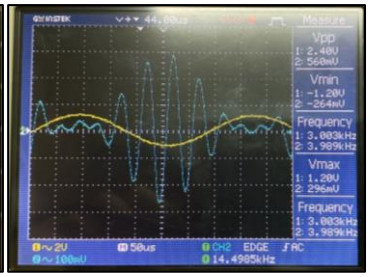
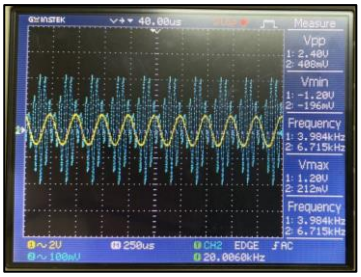
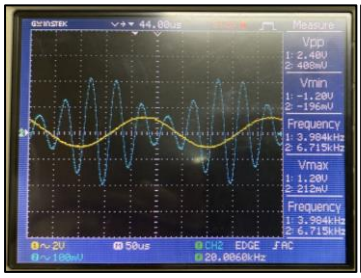
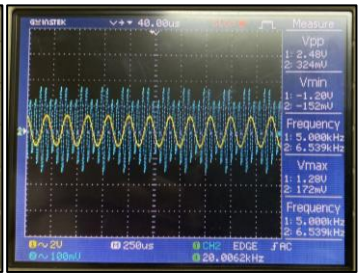
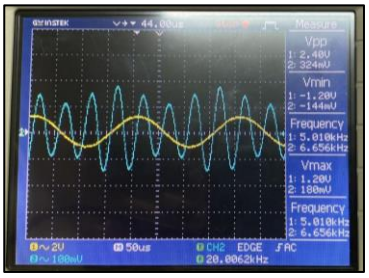
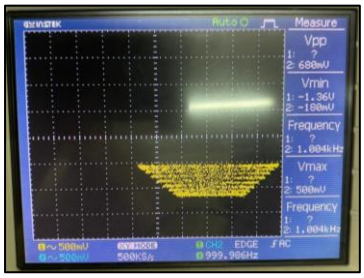
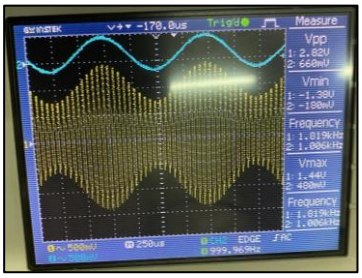
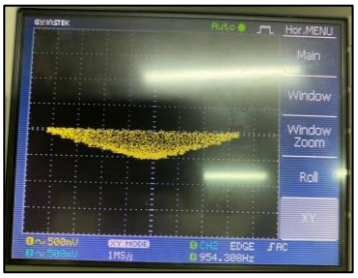
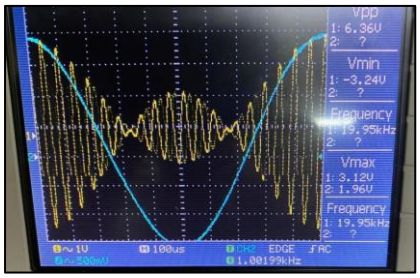
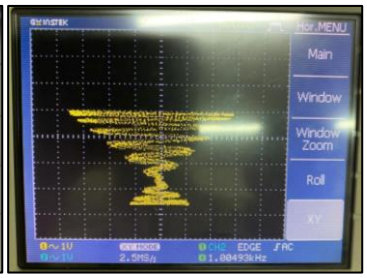
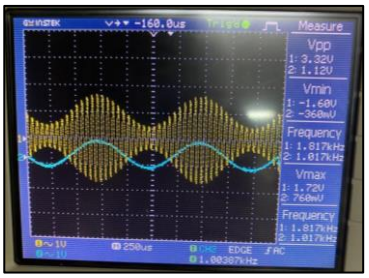
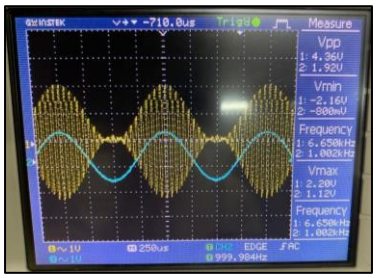
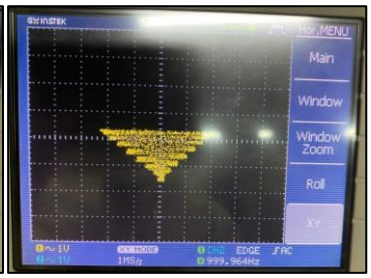
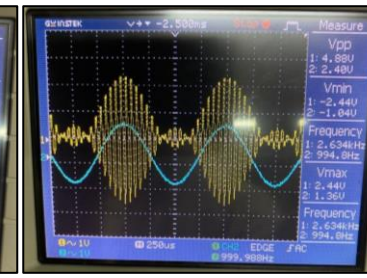
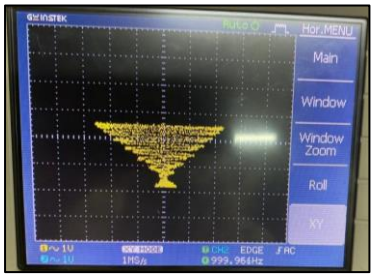
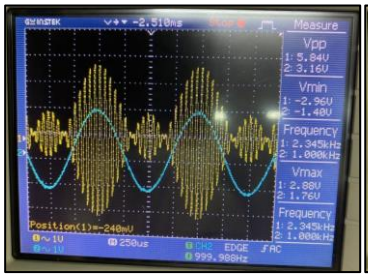
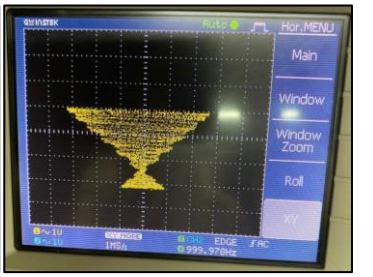


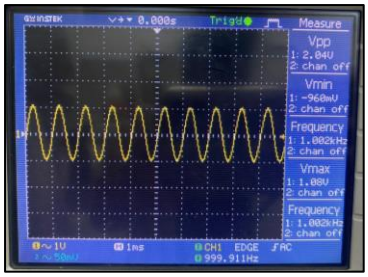
Gambar 5 Bentuk Gelombang Sinyal AM dengan Variasi Harga m

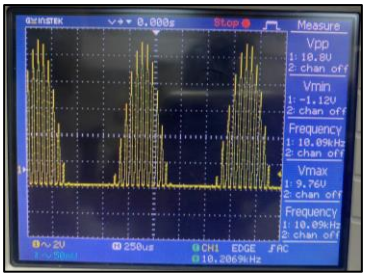


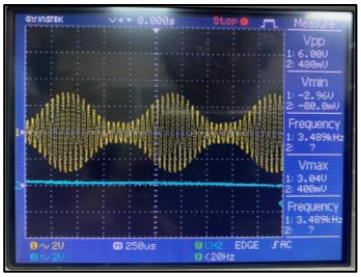













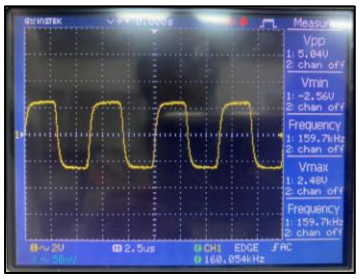




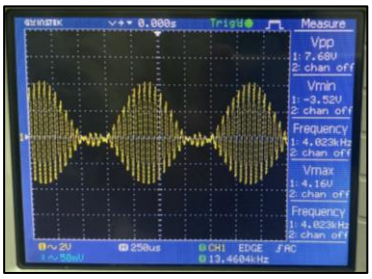



















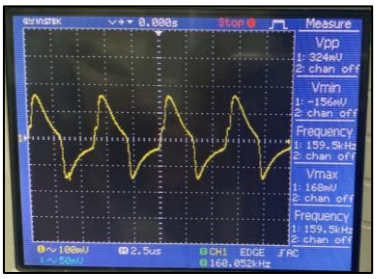

GW INSTEK
0.000s
Trig'd
Measure
Vpp
1: 2.04V
2: chan off
Vmin
1: -960mV
2: chan off
Frequency
1: 1.002kHz
2: chan off
Vmax
1: 1.08V
2: chan off
Frequency
1: 1.002kHz
2: chan off
CH1 EDGE AC
999.911Hz

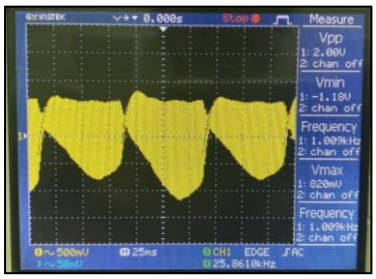

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 10.8V
2: chan off
Vmin
1: -1.12V
2: chan off
Frequency
1: 10.09kHz
2: chan off
Vmax
1: 9.76V
2: chan off
Frequency
1: 10.09kHz
2: chan off
CH1 EDGE AC
10.206kHz

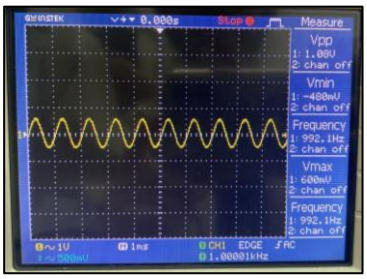

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 6.08V
2: 488mV
Vmin
1: -2.96V
2: -80.0mV
Frequency
1: 3.489kHz
2: ?
Vmax
1: 3.04V
2: 408mV
Frequency
1: 3.489kHz
2: ?
CH2 EDGE AC
<20Hz

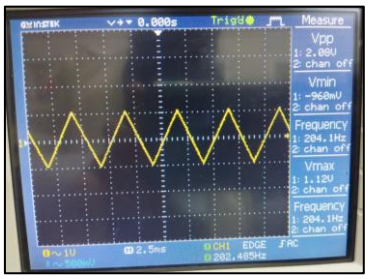

GW INSTEK
Measure
Vpp
1: 5.04V
2: chan off
Vmin
1: -2.56V
2: chan off
Frequency
1: 159.7kHz
2: chan off
Vmax
1: 2.48V
2: chan off
Frequency
1: 159.7kHz
2: chan off
CH1 EDGE AC
160.054kHz

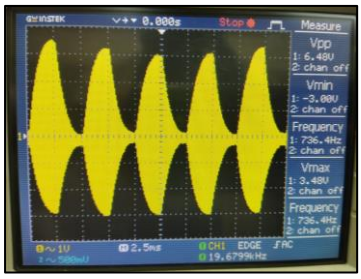

GW INSTEK
0.000s
Trig'd
Measure
Vpp
1: 7.68V
2: chan off
Vmin
1: -3.52V
2: chan off
Frequency
1: 4.023kHz
2: chan off
Vmax
1: 4.16V
2: chan off
Frequency
1: 4.023kHz
2: chan off
CH1 EDGE AC
13.4604kHz

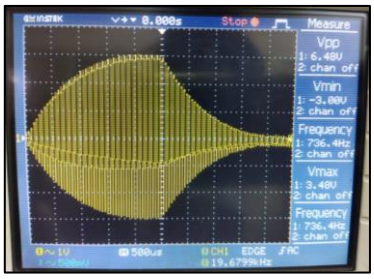

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 324mV
2: chan off
Vmin
1: -156mV
2: chan off
Frequency
1: 159.5kHz
2: chan off
Vmax
1: 168mV
2: chan off
Frequency
1: 159.5kHz
2: chan off
CH1 EDGE AC
160.052kHz

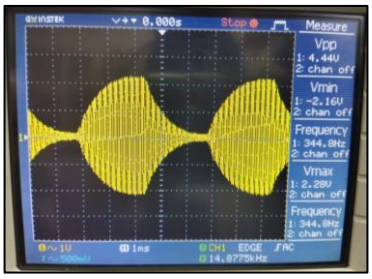

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 2.00V
2: chan off
Vmin
1: -1.18V
2: chan off
Frequency
1: 1.000kHz
2: chan off
Vmax
1: 820mV
2: chan off
Frequency
1: 1.000kHz
2: chan off
CH1 EDGE AC
25.8610kHz

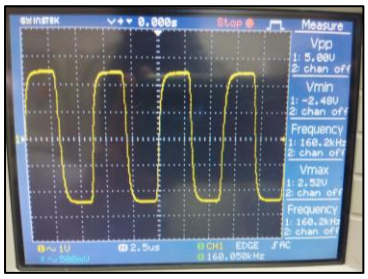

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 1.08V
2: chan off
Vmin
1: -480mV
2: chan off
Frequency
1: 992.1Hz
2: chan off
Vmax
1: 600mV
2: chan off
Frequency
1: 992.1Hz
2: chan off
CH1 EDGE AC
1.00001kHz

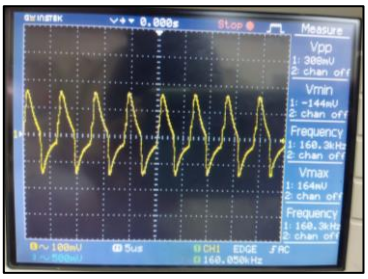

GW INSTEK
0.000s
Trig'd
Measure
Vpp
1: 2.00V
2: chan off
Vmin
1: -960mV
2: chan off
Frequency
1: 204.1Hz
2: chan off
Vmax
1: 1.12V
2: chan off
Frequency
1: 204.1Hz
2: chan off
CH1 EDGE AC
202.485Hz

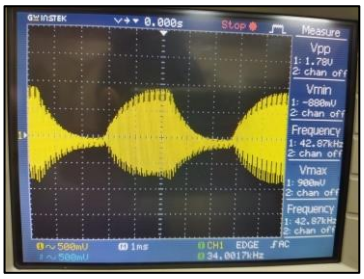

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 6.48V
2: chan off
Vmin
1: -3.00V
2: chan off
Frequency
1: 736.4Hz
2: chan off
Vmax
1: 3.48V
2: chan off
Frequency
1: 736.4Hz
2: chan off
CH1 EDGE AC
19.6799kHz

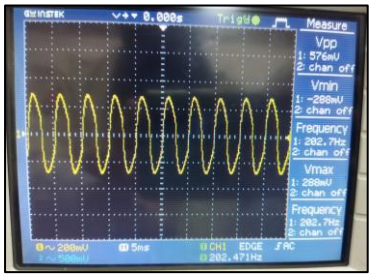

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 6.48V
2: chan off
Vmin
1: -3.00V
2: chan off
Frequency
1: 736.4Hz
2: chan off
Vmax
1: 3.48V
2: chan off
Frequency
1: 736.4Hz
2: chan off
CH1 EDGE AC
19.6799kHz

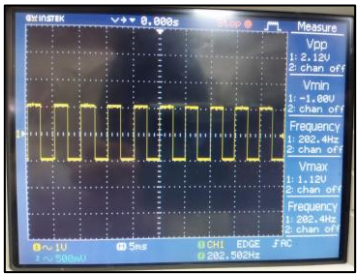

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 4.44V
2: chan off
Vmin
1: -2.16V
2: chan off
Frequency
1: 344.8Hz
2: chan off
Vmax
1: 2.28V
2: chan off
Frequency
1: 344.8Hz
2: chan off
CH1 EDGE AC
14.0775kHz

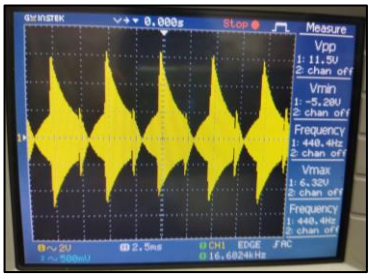

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 5.00V
2: chan off
Vmin
1: -2.48V
2: chan off
Frequency
1: 160.2kHz
2: chan off
Vmax
1: 2.52V
2: chan off
Frequency
1: 160.2kHz
2: chan off
CH1 EDGE AC
160.050kHz

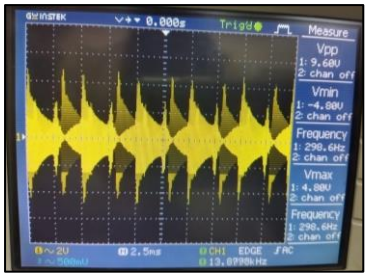
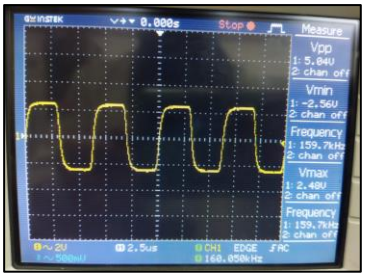
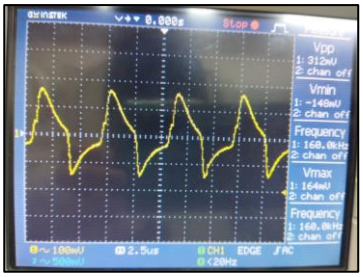
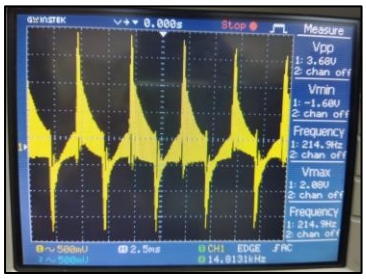
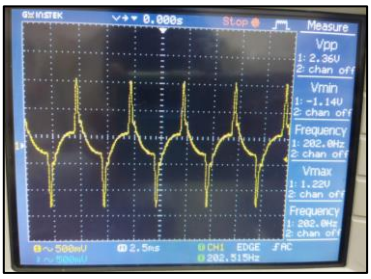

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 309mV
2: chan off
Vmin
1: -144mV
2: chan off
Frequency
1: 160.3kHz
2: chan off
Vmax
1: 164mV
2: chan off
Frequency
1: 160.3kHz
2: chan off
CH1 EDGE AC
160.050kHz

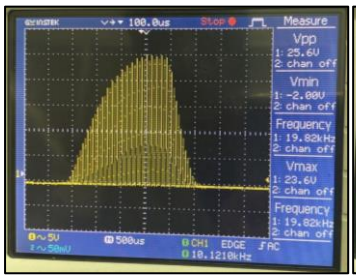
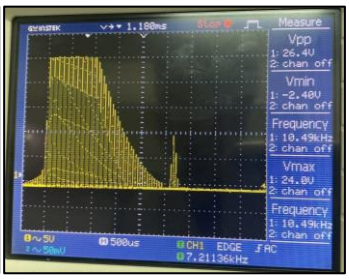
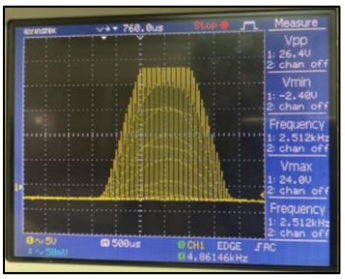

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 1.78V
2: chan off
Vmin
1: -880mV
2: chan off
Frequency
1: 42.87kHz
2: chan off
Vmax
1: 900mV
2: chan off
Frequency
1: 42.87kHz
2: chan off
CH1 EDGE AC
34.0017kHz


GW INSTEK
0.000s
Trig'd
Measure
Vpp
1: 576mV
2: chan off
Vmin
1: -288mV
2: chan off
Frequency
1: 202.7Hz
2: chan off
Vmax
1: 288mV
2: chan off
Frequency
1: 202.7Hz
2: chan off
CH1 EDGE AC
202.471Hz

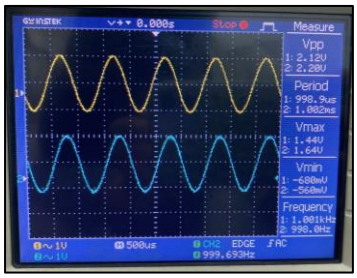

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 2.12V
2: chan off
Vmin
1: -1.00V
2: chan off
Frequency
1: 202.4Hz
2: chan off
Vmax
1: 1.12V
2: chan off
Frequency
1: 202.4Hz
2: chan off
CH1 EDGE AC
202.502Hz

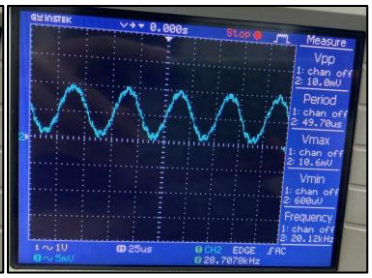
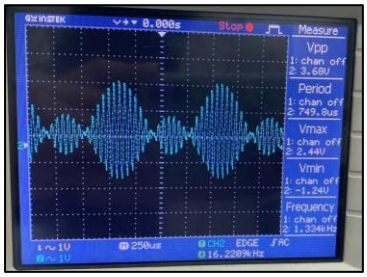
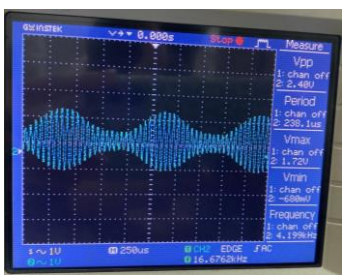
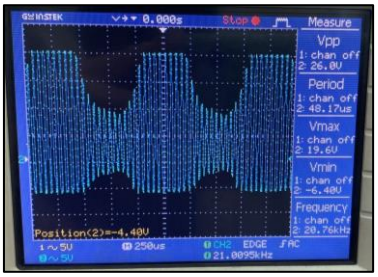
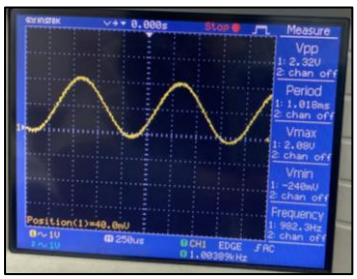
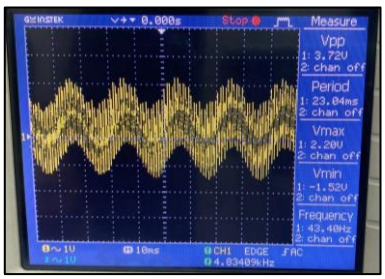
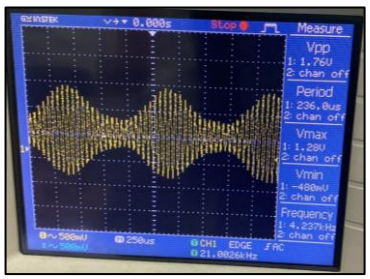

GW INSTEK
0.000s
Stop
Measure
Vpp
1: 11.5V
2: chan off
Vmin
1: -5.20V
2: chan off
Frequency
1: 440.4Hz
2: chan off
Vmax
1: 6.32V
2: chan off
Frequency
1: 440.4Hz
2: chan off
CH1 EDGE AC
16.6024kHz

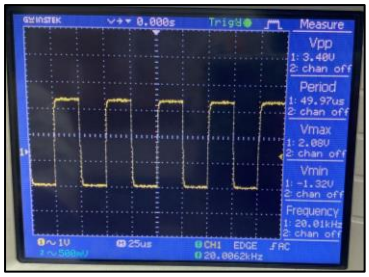






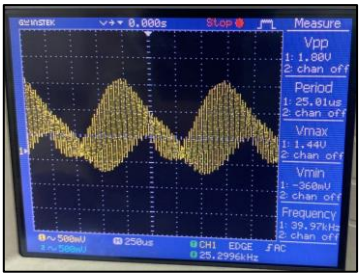




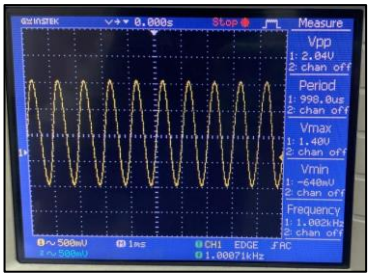
AM SSB SC

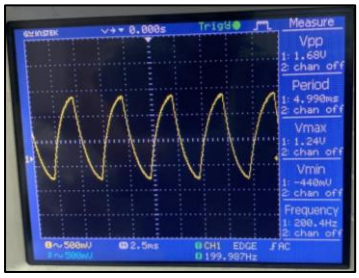


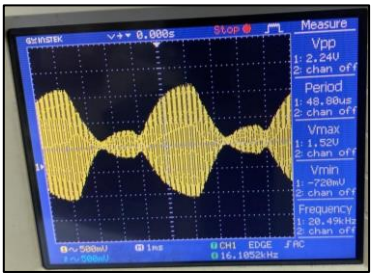








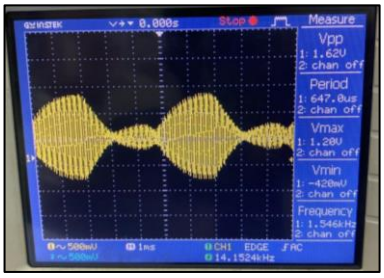

Measure
Vpp
1: 3.48V
Period
1: 49.97us
Vmax
1: 2.08V
Vmin
1: -1.32V
Frequency
1: 20.01kHz

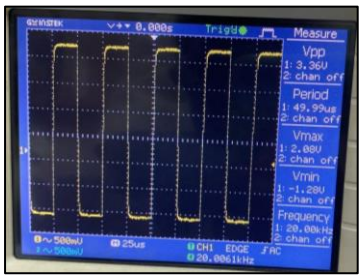

Measure
Vpp
1: 1.88V
Period
1: 25.01us
Vmax
1: 1.44V
Vmin
1: -360mV
Frequency
1: 39.97kHz

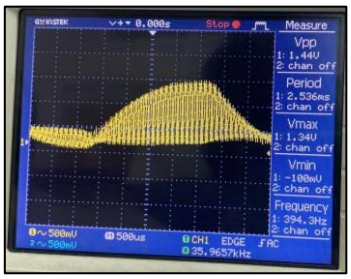

Measure
Vpp
1: 2.04V
Period
1: 998.0us
Vmax
1: 1.40V
Vmin
1: -640mV
Frequency
1: 1.002kHz

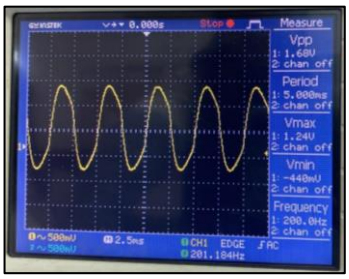

Measure
Vpp
1: 1.68V
Period
1: 4.990ms
Vmax
1: 1.24V
Vmin
1: -440mV
Frequency
1: 200.4Hz

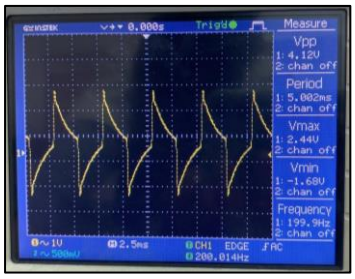

Measure
Vpp
1: 2.24V
Period
1: 40.80us
Vmax
1: 1.52V
Vmin
1: -720mV
Frequency
1: 20.49kHz

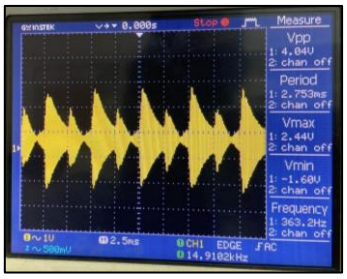

Measure
Vpp
1: 1.62V
Period
1: 647.0us
Vmax
1: 1.20V
Vmin
1: -420mV
Frequency
1: 1.546kHz


Measure
Vpp
1: 3.36V
Period
1: 49.99us
Vmax
1: 2.08V
Vmin
1: -1.28V
Frequency
1: 20.00kHz

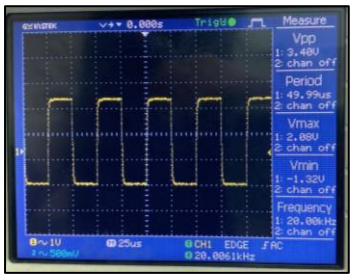

Measure
Vpp
1: 1.44V
Period
1: 2.536ms
Vmax
1: 1.34V
Vmin
1: -100mV
Frequency
1: 394.3Hz

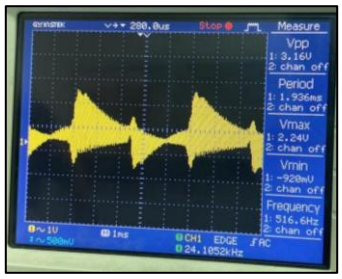

Measure
Vpp
1: 1.68V
Period
1: 5.000ms
Vmax
1: 1.24V
Vmin
1: -440mV
Frequency
1: 200.0Hz


Measure
Vpp
1: 4.12V
Period
1: 5.002ms
Vmax
1: 2.44V
Vmin
1: -1.68V
Frequency
1: 199.9Hz

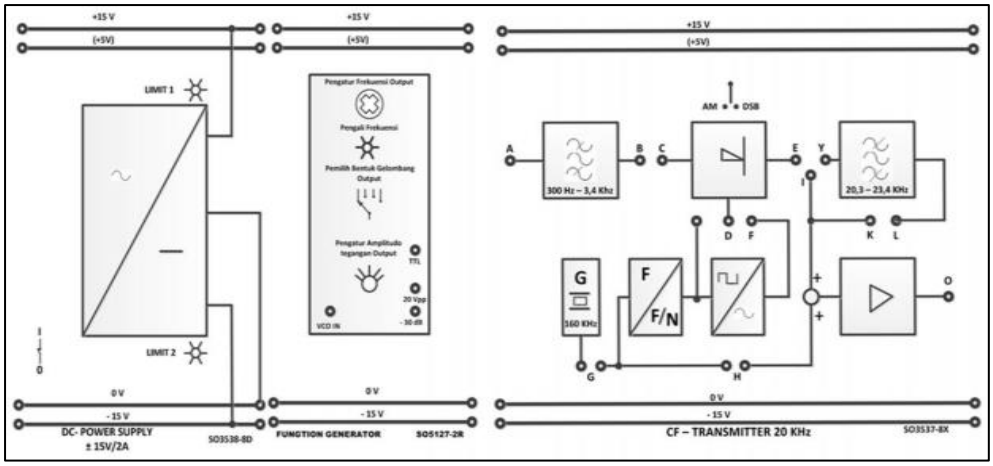

Measure
Vpp
1: 4.04V
Period
1: 2.753ms
Vmax
1: 2.44V
Vmin
1: -1.60V
Frequency
1: 363.2Hz

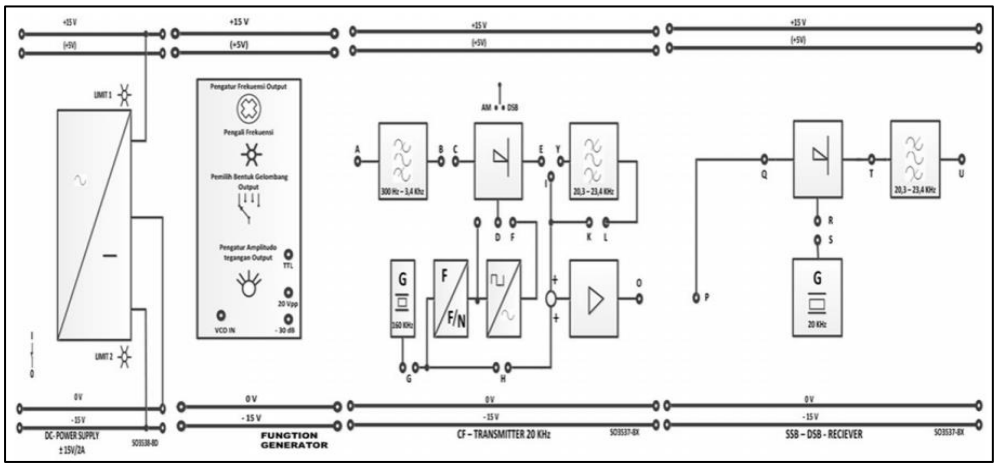

Measure
Vpp
1: 2.88V
Period
1: 451.5us
Vmax
1: 1.80V
Vmin
1: -1.08V
Frequency
1: 2.215kHz


Measure
Vpp
1: 3.48V
Period
1: 49.99us
Vmax
1: 2.08V
Vmin
1: -1.32V
Frequency
1: 20.00kHz


Measure
Vpp
1: 3.16V
Period
1: 1.936ms
Vmax
1: 2.24V
Vmin
1: -920mV
Frequency
1: 516.6Hz


Measure
Vpp
1: 2.22V
Period
1: 4.996ms
Vmax
1: 1.52V
Vmin
1: -700mV
Frequency
1: 200.2Hz


+15 V
(+5V)
LIMIT 1
LIMIT 2
Pengatur Frekuensi Output
Pengali Frekuensi
Pemilih Bentuk Gelombang Output
Pengatur Amplitudo tegangan Output
TTL
20 Vpp
- 30 dB
VCO IN
AM
DSB
300 Hz – 3,4 Khz
20,3 – 23,4 KHz
160 KHz
F/N
0 V
- 15 V
DC- POWER SUPPLY
± 15V/2A
SO3538-8D
FUNGTION GENERATOR
SO5127-2R
CF – TRANSMITTER 20 KHz
SO3537-8X


+15 V
(+5V)
LIMIT 1
LIMIT 2
Pengatur Frekuensi Output
Pengali Frekuensi
Pemilih Bentuk Gelombang Output
Pengatur Amplitudo tegangan Output
TTL
20 Vpp
- 30 dB
VCO IN
AM
DSB
300 Hz – 3,4 Khz
20,3 – 23,4 KHz
160 KHz
20 KHz
F/N
0 V
- 15 V
DC- POWER SUPPLY
± 15V/2A
SO3538-8D
FUNGTION GENERATOR
CF – TRANSMITTER 20 KHz
SO3537-8X
SSB – DSB - RECIEVER
SO3537-8X